# Homework 02: Solutions

REAL VARIABLES I

Fall 2026

Solutions for Homework 02.

## 1. (Images of functions with bounded derivative)

### 1.a

**Remark (Minimal solution).** Since $f(\mathbb{R}) = \bigcup_{N\in\mathbb{N}} f([-N,N])$, by countable subadditivity of $\mathcal{L}_2^*$ it suffices to show that $\mathcal{L}_2^*(f([-N,N])) = 0$ for every $N \in \mathbb{N}$. Fix $N$.
Write $f = (f_1, f_2)$. Fix $n$ and split $[-N,N]$ into $n$ intervals of length $2N/n$:

$$[-N,N] = \bigcup_{k=0}^{n-1} [t_k, t_{k+1}],$$

where $t_k = -N + 2Nk/n$ for $k = 0, \dots, n$. Since $f([-N,N]) = \bigcup_{k=0}^{n-1} f([t_k, t_{k+1}])$, subadditivity of $\mathcal{L}_2^*$ gives

$$\mathcal{L}_2^*(f([-N,N])) \le \sum_{k=0}^{n-1} \mathcal{L}_2^*(f([t_k, t_{k+1}])).$$

By the mean value theorem, $|f_i(s) - f_i(t)| \le \|f'\|_{\sup}|s-t|$ for $i = 1, 2$. Therefore $f([t_k, t_{k+1}])$ lies in the closed square $Q_{n,k}$ of side $4N\|f'\|_{\sup}/n$ centered at $f(t_k)$. By monotonicity of $\mathcal{L}_2^*$ and $\mathcal{L}_2^*$(closed rectangle) $= \mathrm{Vol}_2$,

$$\mathcal{L}_2^*(f([t_k, t_{k+1}])) \le \mathcal{L}_2^*(Q_{n,k}) = \mathrm{Vol}_2(Q_{n,k}) = \Big(\frac{4N\|f'\|_{\sup}}{n}\Big)^2.$$

Hence

$$\mathcal{L}_2^*(f([-N,N])) \le n\Big(\frac{4N\|f'\|_{\sup}}{n}\Big)^2 = \frac{16N^2\|f'\|_{\sup}^2}{n}.$$

Since $n$ is arbitrary, the claim follows by taking $n \to \infty$.

**Remark (Solution).** Write $f = (f_1, f_2)$. The derivative of $f$ is bounded, so $\|f'\|_{\sup} < \infty$. We show that $\mathcal{L}_2^*(f(\mathbb{R})) = 0$. We have $f(\mathbb{R}) = \bigcup_{N\in\mathbb{N}} f([-N,N])$. By countable subadditivity of $\mathcal{L}_2^*$, it suffices to show that $\mathcal{L}_2^*(f([-N,N])) = 0$ for every $N \in \mathbb{N}$. Fix $N$.
**Step 1: $|f_i(s) - f_i(t)| \le \|f'\|_{\sup}|s-t|$ for all $s,t \in \mathbb{R}$ and $i = 1,2$.** For $s = t$ both sides vanish. For $s \ne t$ the mean value theorem gives $\xi$ between $s$ and $t$ with $|f_i(s) - f_i(t)| = |f_i'(\xi)||s-t|$. Moreover $|f_i'(\xi)| \le \|f'(\xi)\| \le \|f'\|_{\sup}$.
**Step 2: $f([-N,N])$ is the union of $n$ sets, each inside a closed square.** Fix $n \ge 1$. Let $t_k = -N + 2Nk/n$ for $k = 0, \dots, n$. Then $[-N,N] = \bigcup_{k=0}^{n-1}[t_k, t_{k+1}]$, and hence $f([-N,N]) = \bigcup_{k=0}^{n-1} f([t_k, t_{k+1}])$. Let $Q_{n,k}$ be the closed square with center $f(t_k)$ and side $4N\|f'\|_{\sup}/n$. Take $t \in [t_k, t_{k+1}]$. Then $|t - t_k| \le 2N/n$. By Step 1, $|f_i(t) - f_i(t_k)| \le 2N\|f'\|_{\sup}/n$ for $i = 1,2$. Therefore $f(t) \in Q_{n,k}$, and so $f([t_k, t_{k+1}]) \subseteq Q_{n,k}$.
**Step 3: $\mathcal{L}_2^*(Q_{n,k}) \le \mathrm{Vol}_2(Q_{n,k})$.** Let $\delta > 0$. The square $Q_{n,k}$ is contained in the open square with the same center and side $4N\|f'\|_{\sup}/n + 2\delta$. Since $\mathcal{L}_2^*$ of an open rectangle is its volume, monotonicity gives $\mathcal{L}_2^*(Q_{n,k}) \le (4N\|f'\|_{\sup}/n + 2\delta)^2$. This holds for every $\delta > 0$. Letting $\delta \to 0$ gives $\mathcal{L}_2^*(Q_{n,k}) \le$

$(4N\|f'\|_{\sup}/n)^2 = \mathrm{Vol}_2(Q_{n,k})$.
**Step 4: conclusion.** By subadditivity and then monotonicity of $\mathcal{L}_2^*$, Steps 2 and 3 give

$$\mathcal{L}_2^*(f([-N,N])) \le \sum_{k=0}^{n-1} \mathcal{L}_2^*(f([t_k,t_{k+1}])) \le \sum_{k=0}^{n-1} \mathcal{L}_2^*(Q_{n,k}) \le n\Big(\frac{4N\|f'\|_{\sup}}{n}\Big)^2 = \frac{16N^2\|f'\|_{\sup}^2}{n}.$$

The left side does not depend on $n$, and $\|f'\|_{\sup}$ and $N$ do not depend on $n$. Since $n$ is arbitrary, the claim follows by taking $n \to \infty$.

## 2. (Equivalent interval definitions of Lebesgue outer measure)

### 2.a

**Remark (Solution).** Let $\mu_o^* = \mathcal{L}_1^*$, $\mu_c^*$, $\mu_d^*$ be the outer measures generated by length on bounded open, bounded closed, and dyadic intervals (in each family $\emptyset$ is admitted with cost 0). We show $\mu_o^* = \mu_c^* = \mu_d^*$. Here $\mu_o^*$ is $\mathcal{L}_1^*$ by definition.
**Lemma.** Let $\mathcal{A}, \mathcal{F} \subseteq 2^X$ contain $\emptyset$, with costs $c_{\mathcal{A}}, c_{\mathcal{F}}$ vanishing at $\emptyset$, and let $\mu_{\mathcal{A}}^*$, $\mu_{\mathcal{F}}^*$ be the generated outer measures. Suppose that for every $A \in \mathcal{A}$ and every $\delta > 0$ there is a countable cover $(F_j)_j \subseteq \mathcal{F}$ of $A$ with $\sum_j c_{\mathcal{F}}(F_j) \le c_{\mathcal{A}}(A) + \delta$. Then $\mu_{\mathcal{F}}^* \le \mu_{\mathcal{A}}^*$.
**Proof.** Let $E \subseteq X$. If $\mu_{\mathcal{A}}^*(E) = +\infty$, the inequality holds because its right side is $+\infty$. Otherwise $\mu_{\mathcal{A}}^*(E) < \infty$; fix $\epsilon > 0$. It suffices to construct a countable cover of $E$ by members of $\mathcal{F}$ of total cost at most $\mu_{\mathcal{A}}^*(E) + 2\epsilon$. Indeed, this gives $\mu_{\mathcal{F}}^*(E) \le \mu_{\mathcal{A}}^*(E) + 2\epsilon$. Since $\epsilon > 0$ is arbitrary, $\mu_{\mathcal{F}}^*(E) \le \mu_{\mathcal{A}}^*(E)$.
Take a cover $(A_i)_{i\in\mathbb{N}} \subseteq \mathcal{A}$ of $E$ with $\sum_{i\in\mathbb{N}} c_{\mathcal{A}}(A_i) \le \mu_{\mathcal{A}}^*(E) + \epsilon$. For each $i$, take a cover $(F_{i,j})_{j\in\mathbb{N}} \subseteq \mathcal{F}$ of $A_i$ with $\sum_{j\in\mathbb{N}} c_{\mathcal{F}}(F_{i,j}) \le c_{\mathcal{A}}(A_i) + \epsilon 2^{-i-1}$. This is possible by the hypothesis with $\delta = \epsilon 2^{-i-1}$. A finite cover is padded with $\emptyset \in \mathcal{F}$, which costs 0. The family $(F_{i,j})_{(i,j)\in\mathbb{N}^2}$ lies in $\mathcal{F}$ and is countable. Enumerate $\mathbb{N}^2$ by a bijection with $\mathbb{N}$. The total cost is a sum of nonnegative terms, so it does not depend on the enumeration. The family covers $E$, because the $A_i$ cover $E$ and the $F_{i,j}$ cover $A_i$. Its total cost is at most $\sum_{i\in\mathbb{N}}(c_{\mathcal{A}}(A_i) + \epsilon 2^{-i-1})$. This is at most $\mu_{\mathcal{A}}^*(E) + 2\epsilon$, since $\sum_{i\in\mathbb{N}} c_{\mathcal{A}}(A_i) \le \mu_{\mathcal{A}}^*(E) + \epsilon$ and $\sum_{i\in\mathbb{N}} \epsilon 2^{-i-1} = \epsilon$.
By the Lemma, each of (i)–(iv) follows from its hypothesis. For every interval $A$ of the first family and every $\delta > 0$ we exhibit a countable cover of $A$ by intervals of the second family of total length at most the length of $A$ plus $\delta$. For $A = \emptyset$ the cover $(\emptyset)$ has total cost 0. Hence we may take $A \neq \emptyset$. In each item, $\mathcal{A}$ is the family generating the outer measure on the right of the inequality and $\mathcal{F}$ the family generating the one on the left. The four inequalities give $\mu_c^* \le \mu_o^* \le \mu_c^*$ and $\mu_d^* \le \mu_o^* \le \mu_d^*$. Hence $\mu_o^* = \mu_c^* = \mu_d^*$.
**(i)** $\mu_c^* \le \mu_o^*$**.** The open interval $(a,b)$ is covered by the closed interval $[a,b]$ of the same length.
**(ii)** $\mu_o^* \le \mu_c^*$**.** A closed interval $[a,b]$ is covered by $(a-\delta/2, b+\delta/2)$, of length $b-a+\delta$.
**(iii)** $\mu_o^* \le \mu_d^*$**.** A dyadic interval $[k2^{-m}, (k+1)2^{-m})$ is covered by $(k2^{-m} - \delta/2, (k+1)2^{-m} + \delta/2)$, of length $2^{-m} + \delta$.
**(iv)** $\mu_d^* \le \mu_o^*$**.** Let $A = (a,b)$ with $a < b$ and let $\delta > 0$. Choose $m \in \mathbb{N}$ with $2 \cdot 2^{-m} \le \delta$. Then $m$ depends only on $\delta$, not on $a, b$. The dyadic intervals $[k2^{-m}, (k+1)2^{-m})$, $k \in \mathbb{Z}$, partition $\mathbb{R}$. Therefore those meeting $(a,b)$ cover $(a,b)$. Such an interval meets $(a,b)$ exactly when $a2^m - 1 < k < b2^m$. This is an open interval of length $(b-a)2^m + 1$. Hence it contains at most $(b-a)2^m + 2$ integers. Their total length is therefore at most $b - a + 2 \cdot 2^{-m} \le b - a + \delta$.

## 3. (Finite interval covers and outer Jordan content)

### 3.a

**Remark (Minimal solution).** For every $\delta > 0$, $(0,1) \subseteq [0,1] \subseteq (-\delta, 1+\delta)$. By monotonicity and $\mathcal{L}_1^*$(open interval) = length,

$$1 = \mathcal{L}_1^*((0,1)) \leq \mathcal{L}_1^*([0,1]) \leq \mathcal{L}_1^*((-\delta, 1+\delta)) = 1 + 2\delta.$$

Letting $\delta \to 0$ gives $\mathcal{L}_1^*([0,1]) = 1$. By monotonicity, $\mathcal{L}_1^*([0,1] \setminus \mathbb{Q}) \leq \mathcal{L}_1^*([0,1]) = 1$. Since $[0,1] \cap \mathbb{Q}$ is countable, $\mathcal{L}_1^*([0,1] \cap \mathbb{Q}) = 0$ by the nullity of countable sets. By subadditivity,

$$1 = \mathcal{L}_1^*([0,1]) \leq \mathcal{L}_1^*([0,1] \setminus \mathbb{Q}) + \mathcal{L}_1^*([0,1] \cap \mathbb{Q}) = \mathcal{L}_1^*([0,1] \setminus \mathbb{Q}).$$

**Remark (Solution).** We show that $\mathcal{L}_1^*([0,1] \setminus \mathbb{Q}) = 1$.
**Step 1:** $\mathcal{L}_1^*([0,1]) = 1$**.** Fix $\delta > 0$; we show $1 \leq \mathcal{L}_1^*([0,1]) \leq 1 + 2\delta$. We have $(0,1) \subseteq [0,1] \subseteq (-\delta, 1+\delta)$. Since $\mathcal{L}_1^*$ of an open interval is its length and $\mathcal{L}_1^*$ is monotone, $1 \leq \mathcal{L}_1^*([0,1]) \leq 1 + 2\delta$. This holds for every $\delta > 0$, so letting $\delta \to 0$ gives the claim.
**Step 2: conclusion.** We show that $\mathcal{L}_1^*([0,1] \setminus \mathbb{Q}) \leq 1$ and $\mathcal{L}_1^*([0,1] \setminus \mathbb{Q}) \geq 1$. The set $[0,1] \cap \mathbb{Q}$ is countable, hence $\mathcal{L}_1^*$-null. Monotonicity gives $\mathcal{L}_1^*([0,1] \setminus \mathbb{Q}) \leq \mathcal{L}_1^*([0,1]) = 1$. Subadditivity gives

$$1 = \mathcal{L}_1^*([0,1]) \leq \mathcal{L}_1^*([0,1] \setminus \mathbb{Q}) + \mathcal{L}_1^*([0,1] \cap \mathbb{Q}) = \mathcal{L}_1^*([0,1] \setminus \mathbb{Q}).$$

**3.b**

**Remark (Solution).** We show that $J(E) = J(\bar{E})$ for every $E \subseteq \mathbb{R}$ (with $\inf \emptyset = +\infty$), and that $E = [0,1] \cap \mathbb{Q}$ satisfies $J(E) = 1$ and $\mathcal{L}_1^*(E) = 0$.
**Step 1:** $J(E) = J(\bar{E})$**.** A finite union $U$ of closed intervals is closed. Hence $E \subseteq U$ implies $\bar{E} \subseteq U$. Conversely, $\bar{E} \subseteq U$ implies $E \subseteq U$. Therefore a finite collection of closed intervals has union containing $E$ if and only if its union contains $\bar{E}$. Hence, according to the definition of $J$ as an infimum over such collections, it holds that $J(E) = J(\bar{E})$.
**Step 2: if** $[c,d] \subseteq \bigcup_{n<N}[a_n, b_n]$**, then** $d - c \leq \sum_{n<N}(b_n - a_n)$**.** If $c = d$, the claim holds because the right side is nonnegative. Let $c < d$ and fix $\delta > 0$. It suffices to show that $d - c \leq \sum_{n<N}(b_n - a_n) + 2N\delta$ for every $\delta > 0$. The intervals, hence $N$, are fixed before $\delta$. Therefore letting $\delta \to 0$ gives the claim. We have $(c,d) \subseteq [c,d] \subseteq \bigcup_{n<N}(a_n - \delta, b_n + \delta)$. Hence, by monotonicity, subadditivity, and $\mathcal{L}_1^*$(open interval) = length,

$$d - c \leq \sum_{n<N}(b_n - a_n + 2\delta) = \sum_{n<N}(b_n - a_n) + 2N\delta.$$

**Step 3:** $J([0,1]) = 1$**.** The cover $\{[0,1]\}$ gives $J([0,1]) \leq 1$, and Step 2 gives $J([0,1]) \geq 1$.
**Step 4: the example.** Let $E = [0,1] \cap \mathbb{Q}$. It is countable, so $\mathcal{L}_1^*(E) = 0$. Since $\mathbb{Q}$ is dense and $[0,1]$ is closed, $\bar{E} = [0,1]$. By Steps 1 and 3, $J(E) = J([0,1]) = 1 \neq 0 = \mathcal{L}_1^*(E)$.

**Remark (Discussion).** Finite covers see only the closure of $E$. Hence they cannot detect that a dense countable set is negligible. Countable covers can, by shrinking the $j$-th interval geometrically.